

Beyond Variance: A Monte Carlo Analysis of Downside Risk in Portfolio Selection

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Abstract

This thesis investigates portfolio selection under downside risk and compares it with the traditional mean–variance framework. While variance treats positive and negative deviations symmetrically, downside risk focuses specifically on returns falling below a target level, set in this study equal to the risk-free return. The theoretical analysis develops the main properties of the downside-risk measure, the downside-cosemivariance matrix, downside-efficient portfolios, the efficient frontier in the presence and absence of a risk-free asset, and the role of the Sortino ratio.

The empirical analysis is conducted through Monte Carlo simulation in MATLAB. A market with five risky assets and one risk-free asset is simulated under three settings: a pure lognormal scenario, a moderate-default scenario (Jarrow), and a more severe “dangerous” scenario. The simulation uses 100,000 observations and introduces increasing probabilities of default to generate asymmetric return distributions and more pronounced left-tail risk. Mean–variance efficient portfolios are compared with downside-efficient portfolios, including a comparison at equal expected return.

The results show that, in the absence of default, the two approaches produce very similar portfolio allocations and risk profiles. As default probabilities increase, however, the downside-efficient portfolio reallocates more strongly toward assets with lower default risk and reduces exposure to assets characterized by more severe downside outcomes. At equal expected return, this approach provides better protection in the left tail of the return distribution: the advantage in terms of 1% Expected Shortfall becomes progressively larger as default intensity increases, while the 1% quantile and other extreme-loss indicators also improve.

Robustness and sensitivity analyses support the numerical reliability of the results. The downside-cosemivariance matrix is statistically more difficult to estimate than the covariance matrix—its element-wise estimation error is approximately seven times larger in the benchmark experiment—yet the final portfolio risk measures remain stable across Monte Carlo replications. Further tests on sample size, default intensity and random seed confirm that the main findings are structural rather than driven by a particular simulation draw.

Overall, the analysis shows that the choice of risk measure becomes particularly relevant when returns are asymmetric and exposed to extreme events. In relatively regular environments, mean–variance and downside-efficient portfolio selection lead to comparable results; under increasing default risk, a downside-oriented criterion provides a more targeted representation of economically relevant losses and a stronger protection against severe tail outcomes.